

Air Quality & Weather Health Report - Nanded

1. Summary

This report provides a comprehensive overview of the air quality and weather conditions in Nanded as of the reporting time. The Predicted AQI is 48, indicating moderate air quality. While generally acceptable, some pollutants are present at elevated levels requiring monitoring. The weather is warm and relatively humid with a moderate breeze. Specific recommendations are provided for sensitive groups.

2. Gaseous Pollutants (NO, NO2, NOx, SO2, CO, PM2.5, PM10)

The following table details the concentrations of various gaseous pollutants in Nanded:

Pollutant	Concentration	Units
NO	1	ppb
NO2	4	ppb
NOx (ppb)	3	ppb
SO2	3	ppb
CO	0	mg/m³
PM2.5	13	µg/m³
PM10	19	µg/m³
NH3	16	ppb

While CO levels are low, PM2.5 and PM10 levels are moderately elevated, contributing to the overall AQI. Further investigation into the sources of these particulate pollutants is warranted.

3. Ozone Levels

The ozone concentration in Nanded is 18 ppb. While not excessively high, prolonged exposure to elevated ozone levels can still impact respiratory health. It's important to note that ozone formation is often related to sunlight and other pollutants, suggesting a potential link to the higher PM levels.

4. Weather Overview

Nanded is currently experiencing warm weather with a temperature of 26°C (temperature_2m) and 31°C (AT). The relative humidity is 76% (relative_humidity_2m) and 64% (RH). Wind speed is moderate at 2 m/s (WS) and 14 km/h (wind_speed_10m), blowing from a direction of 251° (WD) and 285° (wind_direction_10m). There is no rainfall currently (RF, rain).

5. Health Risks Summary

The moderate AQI of 48 suggests a low to moderate health risk for the general population. However, individuals with pre-existing respiratory or cardiovascular conditions may experience increased symptoms. Children, the elderly, and pregnant women are particularly vulnerable and should take precautions.

6. Cause of Pollution (Predicted)

The moderately elevated levels of PM2.5 and PM10 in Nanded suggest potential contributions from various sources. These could include vehicular emissions, industrial activities, construction dust, and agricultural practices. The relatively low wind speed might also contribute to pollutant accumulation. Further investigation using more detailed pollution source tracking would be needed to pinpoint the exact causes with certainty.

7. Safety Recommendations

For the general population, it is advisable to limit strenuous outdoor activities during periods of higher pollution. Stay hydrated, and consider wearing a mask if engaging in outdoor activities. For those with respiratory sensitivities, it is recommended to remain indoors as much as possible during periods of high pollution.

8. Tips for Sensitive Groups (Children, Elderly, Pregnant Women, Respiratory Patients)

Children, the elderly, pregnant women, and individuals with respiratory conditions should take extra precautions during periods of moderate air quality. Limit outdoor time, especially during peak pollution hours. Use air purifiers indoors and ensure adequate ventilation. Consult a doctor if experiencing respiratory symptoms.